CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 5 of 10)

This is Set 5 of the Half-Yearly Predicted Paper series, precisely aligned with the official CBSE blueprint for the 2026-27 academic session. This paper strictly tests the ~80% half-yearly syllabus, completely excluding Linear Programming and Probability. Every question is highly probable, based on recent competency-focused trends and frequently tested PYQs.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let $A = \{1, 2, 3\}$. Find the number of relations containing $(1, 2)$ and $(1, 3)$ which are reflexive and symmetric but not transitive. * **Chapter:** Relations and Functions * **Confidence: Very High**

Q2. Find the domain of the inverse trigonometric function $f(x) = \cos^{-1}(2x - 1)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: High**

Q3. If $A = [a_{ij}]$ is a 3×3 matrix such that $a_{ij} = i^2 - j^2$, then A is which type of matrix? * **Chapter:** Matrices * **Confidence: Very High**

Q4. Without expanding, find the value of the determinant $\begin{vmatrix} \sin^2 x & \cos^2 x & 1 \\ \cos^2 x & \sin^2 x & 1 \\ -10 & 12 & 2 \end{vmatrix}$. * **Chapter:** Determinants * **Confidence: High**

Q5. What is the derivative of $e^{\sin^{-1} x}$ with respect to x? * **Chapter:** Continuity and Differentiability * **Confidence: High**

Q6. For the curve $y = x^3 - 3x + 2$, find the points on the curve where the tangent is completely horizontal (parallel to the x-axis). * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q7. Evaluate the indefinite integral: $\int \frac{1-\sin x}{\cos^2 x}\, dx$. * **Chapter:** Integrals * **Confidence: High**

Q8. Evaluate the definite integral: $\int_{-\pi/2}^{\pi/2} \sin^5 x \cos x\, dx$. * **Chapter:** Integrals * **Confidence: Very High**

Q9. State the exact degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{dy}{dx}\right)^5 = x^3$. * **Chapter:** Differential Equations * **Confidence: Very High**

Q10. Write the general solution of the differential equation $\frac{dy}{dx} = \sec x$. * **Chapter:** Differential Equations * **Confidence: Medium**

Q11. Find a unit vector in the direction of the vector $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$. * **Chapter:** Vector Algebra * **Confidence: High**

Q12. What is the area of a parallelogram having adjacent sides represented by the vectors $\hat{i} - \hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} - \hat{k}$? * **Chapter:** Vector Algebra * **Confidence: Very High**

Q13. Write the direction cosines of a straight line that is strictly parallel to the y-axis. * **Chapter:** Three-Dimensional Geometry * **Confidence: High**

Q14. Write the Cartesian equation of a line passing through $(1, -2, 3)$ with direction ratios proportional to $2, -1, 4$. * **Chapter:** Three-Dimensional Geometry * **Confidence: High**

Q15. Let R be the relation in the set of natural numbers $\mathbb{N}$ given by $R = \{(a, b) : a = b - 2, b > 6\}$. Does the pair $(6, 8)$ belong to relation R? * **Chapter:** Relations and Functions * **Confidence: Medium**

Q16. Find the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: Very High**

Q17. If A is a square matrix such that $A^2 = I$ (Identity Matrix), what is A^{-1} equal to? * **Chapter:** Matrices * **Confidence: High**

Q18. The radius of a spherical soap bubble is increasing at the rate of $1/2\ \mathrm{cm/s}$. Find the rate of change of its volume when the radius is $1\ \mathrm{cm}$. * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q19. (Assertion-Reason): Assertion (A): The function $f(x) = \frac{1}{x-2}$ is continuous at $x = 2$. **Reason (R):** A rational function is continuous at all points in its domain. Choose the correct option. * **Chapter:** Continuity and Differentiability * **Confidence: High**

Q20. (Assertion-Reason): Assertion (A): The vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ is equally inclined to all three coordinate axes. **Reason (R):** The direction cosines of the vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ are equal. Choose the correct option. * **Chapter:** Vector Algebra * **Confidence: Very High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Find the exact numerical value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: Very High**

Q22. Find the matrix X such that $2A + 3X = 5B$, given $A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & 2 \end{bmatrix}$. * **Chapter:** Matrices * **Confidence: High**

Q23. Differentiate $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to x, assuming x lies in the principal domain. * **Chapter:** Continuity and Differentiability * **Confidence: Very High**

Q24. Solve the differential equation using variable separation: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$. * **Chapter:** Differential Equations * **Confidence: High**

Q25. Find a vector of magnitude 5 units that is perpendicular to both $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$. * **Chapter:** Vector Algebra * **Confidence: Very High**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Let $A = \mathbb{R} - \{2\}$ and $B = \mathbb{R} - \{1\}$. If $f : A \to B$ is a function defined by $f(x) = \frac{x-1}{x-2}$, show algebraically that f is one-one and onto. * **Chapter:** Relations and Functions * **Confidence: Very High**

Q27. Using the properties of determinants, prove that: $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$. * **Chapter:** Determinants * **Confidence: Very High**

Q28. Differentiate the implicit-exponential function $x^y = e^{x-y}$ with respect to x and prove that $\frac{dy}{dx} = \frac{\log_e x}{(1+\log_e x)^2}$. * **Chapter:** Continuity and Differentiability * **Confidence: Very High**

Q29. Evaluate the indefinite integral: $\int \frac{x+3}{\sqrt{5-4x-x^2}}\,dx$. * **Chapter:** Integrals * **Confidence: High**

Q30. Solve the linear differential equation: $(x \log x)\frac{dy}{dx} + y = \frac{2}{x}\log x$. * **Chapter:** Differential Equations * **Confidence: Very High**

Q31. Find the coordinates of the point where the line passing through the points $A(3, 4, 1)$ and $B(5, 1, 6)$ crosses the xy-plane. * **Chapter:** Three-Dimensional Geometry * **Confidence: High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Solve the following system of linear equations using the matrix inverse method: $x - y + 2z = 1$ $2y - 3z = 1$ $3x - 2y + 4z = 2$ * **Chapter:** Matrices and Determinants * **Confidence: Very High**

Q33. Show mathematically that the altitude of a right circular cylinder of maximum volume that can be inscribed in a given sphere of radius R is $\frac{2R}{\sqrt{3}}$. * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q34. Evaluate the following definite integral using properties of limits: $\int_0^{\pi/4} \log(1 + \tan x)\,dx$. * **Chapter:** Integrals * **Confidence: Very High**

Q35. Sketch the region and find the area bounded by the curve $y = x^2 + 2$, the line $y = x$, and the ordinates $x = 0$ and $x = 3$. * **Chapter:** Application of Integrals * **Confidence: High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Financial Investment) A cooperative trust fund has ₹ $30,000$ that must be invested in two different types of bonds. The first bond pays 5% interest per year, and the second bond pays 7% interest per year. The trust fund must obtain an annual total interest of exactly ₹ $1,800$. Let x be the amount invested in the first bond and y be the amount invested in the second bond. *(i)* Formulate a 1×2 matrix for the investment amounts and a 2×1 matrix for the interest rates. *(1 Mark) (ii)* Set up the matrix multiplication equation to represent the total annual interest. *(1 Mark) (iii)* Using matrix multiplication, calculate the exact amount invested in each of the two bonds. *(2 Marks)* * **Chapter:** Matrices * **Confidence: High**

Q37. Case Study 2: Application of Derivatives (Wire Cutting & Optimization) A manufacturing firm has a piece of industrial wire exactly 28 m in length. It is to be cut into two pieces. One of the pieces is bent to form a square, and the other piece is bent to form a circle. Let the length of the wire piece used for the square be $4x$ (so the side of the square is x). *(i)* Express the combined total area A of the square and the circle strictly in terms of x. *(1 Mark) (ii)* Find the first derivative $\frac{dA}{dx}$ of the area function. *(1 Mark) (iii)* Find the exact length of the piece that should be formed into the square so that the combined area is mathematically minimum. *(2 Marks)* * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q38. Case Study 3: 3D Geometry (Submarine Paths) Two military submarines are traversing deep in the ocean. Using a 3D coordinate mapping system, the path of Submarine 1 is determined by the straight-line vector equation: $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$. The path of Submarine 2 is given by the line: $\vec{r} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(2\hat{i} + 3\hat{j} + \hat{k})$. *(i)* Write the Cartesian equation representing the path of Submarine 1. *(1 Mark) (ii)* Check mathematically if the direction vectors of the two submarine paths are orthogonal (perpendicular). *(1 Mark) (iii)* Find the exact shortest distance between the two submarine paths to determine how close they will pass each other. *(2 Marks)* * **Chapter:** Three-Dimensional Geometry * **Confidence: Very High**

Answer Key

Q1. 1 relation. The set must contain $(1,1),(2,2),(3,3)$ for reflexivity. To be symmetric and contain $(1,2)$ and $(1,3)$, it must contain $(2,1)$ and $(3,1)$. Thus, $R=\{(1,1),(2,2),(3,3),(1,2),(2,1),(1,3),(3,1)\}$. This relation is not transitive because $(2,1)\in R$ and $(1,3)\in R$, but $(2,3)\notin R$. **Q2.** We require $-1\le 2x-1\le 1 \implies 0\le 2x\le 2 \implies 0\le x\le 1$. Domain is $[1]$. **Q3.** Since $a_{ij}=i^2-j^2$, we have $a_{ji}=j^2-i^2=-(i^2-j^2)=-a_{ij}$, and $a_{ii}=0$. Therefore, A is a Skew-Symmetric matrix. **Q4.** By applying the column operation $C_1\to C_1+C_2$, the first column becomes $\sin^2 x+\cos^2 x=1$. The determinant has two identical columns (C_1 and C_3 are both $1,1,2$ if we scale C_3, wait... C_1 becomes $1,1,2$, and C_3 is $1,1,2$). Since two columns are identical, the determinant is 0. **Q5.** By chain rule: $\frac{d}{dx}(e^{\sin^{-1}x})=e^{\sin^{-1}x}\cdot\frac{1}{\sqrt{1-x^2}}$. **Q6.** Horizontal tangent means $\frac{dy}{dx}=0$. $\frac{dy}{dx}=3x^2-3=0 \implies x^2=1 \implies x=\pm1$. At $x=1,y=0$. At $x=-1,y=4$. Points are $(1,0)$ and $(-1,4)$. **Q7.** $\int\left(\frac{1}{\cos^2 x}-\frac{\sin x}{\cos^2 x}\right)dx=\int(\sec^2 x-\sec x\tan x)dx=\tan x-\sec x+C$. **Q8.** Let $f(x)=\sin^5 x\cos x$. Since $f(-x)=(-\sin x)^5\cos(-x)=-\sin^5 x\cos x=-f(x)$, it is an odd function. Thus, the integral is 0. **Q9.** The highest order derivative is $\frac{d^3y}{dx^3}$. Its power is 2. Hence, the degree is 2. **Q10.** $\int dy=\int\sec x\,dx \implies y=\log|\sec x+\tan x|+C$. **Q11.** $|\vec a|=\sqrt{2^2+(-3)^2+6^2}=\sqrt{4+9+36}=\sqrt{49}=7$. Unit vector $\hat a=\frac{2}{7}\hat i-\frac{3}{7}\hat j+\frac{6}{7}\hat k$. **Q12.** Area $=|\vec a\times\vec b|$. $\vec a\times\vec b=\begin{vmatrix}\hat i&\hat j&\hat k\\1&-1&1\\2&1&-1\end{vmatrix}=\hat i(1-1)-\hat j(-1-2)+\hat k(1+2)=3\hat j+3\hat k$. Magnitude $=\sqrt{0+9+9}=\sqrt{18}=3\sqrt2$ sq units. **Q13.** The direction cosines of a line parallel to the y-axis are $0,1,0$ (or $0,-1,0$). **Q14.** $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z-3}{4}$. **Q15.** The relation specifies $a=b-2$ and $b>6$. For the pair $(6,8)$, $b=8>6$ is satisfied, and $a=8-2=6$ is satisfied. Yes, $(6,8)\in R$. **Q16.** $\sin^{-1}(-\frac12)=-\sin^{-1}(\frac12)=-\frac{\pi}{6}$. **Q17.** Pre-multiplying both sides of $A^2=I$ by A^{-1}, we get $A^{-1}A^2=A^{-1}I \implies A=A^{-1}$. Thus, $A^{-1}=A$. **Q18.** $V=\frac43\pi r^3 \implies \frac{dV}{dt}=4\pi r^2\frac{dr}{dt}$. At $r=1$ and $\frac{dr}{dt}=1/2$, $\frac{dV}{dt}=4\pi(1)^2(1/2)=2\pi\ \text{cm}^3/\text{s}$. **Q19.** Option D (Assertion is false, Reason is true). The domain of $f(x)$ is $\mathbb{R}-\{2\}$. Since $x=2$ is not in the domain, the function is not continuous at $x=2$. **Q20.** Option A (Both A and R are true, and R is the correct explanation of A). DCs are $\frac{1}{\sqrt3},\frac{1}{\sqrt3},\frac{1}{\sqrt3}$, which implies $\cos\alpha=\cos\beta=\cos\gamma$.

Q21. $\tan^{-1}(1)=\frac{\pi}{4}$. $\cos^{-1}(-\frac12)=\frac{2\pi}{3}$. $\sin^{-1}(-\frac12)=-\frac{\pi}{6}$. Sum $=\frac{\pi}{4}+\frac{2\pi}{3}-\frac{\pi}{6}=\frac{3\pi+8\pi-2\pi}{12}=\frac{9\pi}{12}=\frac{3\pi}{4}$. **Q22.** $3X=5B-2A=\begin{bmatrix}10&-10\\20&10\end{bmatrix}-\begin{bmatrix}16&0\\8&-4\end{bmatrix}=\begin{bmatrix}-6&-10\\12&14\end{bmatrix}$. Thus, $X=\begin{bmatrix}-2&-10/3\\4&14/3\end{bmatrix}$. **Q23.** Let $x=\tan\theta$. Then $y=\sin^{-1}(\frac{2\tan\theta}{1+\tan^2\theta})=\sin^{-1}(\sin2\theta)=2\theta=2\tan^{-1}x$. Therefore, $\frac{dy}{dx}=\frac{2}{1+x^2}$. **Q24.** Separating variables: $\frac{dy}{1+y^2}=\frac{dx}{1+x^2}$. Integrating both sides: $\tan^{-1}y=\tan^{-1}x+C$. **Q25.** Cross product $\vec n=\vec a\times\vec b=\begin{vmatrix}\hat i&\hat j&\hat k\\1&1&0\\0&1&1\end{vmatrix}=\hat i(1)-\hat j(1)+\hat k(1)=\hat i-\hat j+\hat k$. Magnitude $|\vec n|=\sqrt3$. Required vector $=\pm5\frac{(\hat i-\hat j+\hat k)}{\sqrt3}$.

Q26. One-one: Let $f(x_1)=f(x_2) \implies \frac{x_1-1}{x_1-2}=\frac{x_2-1}{x_2-2} \implies x_1x_2-2x_1-x_2+2=x_1x_2-x_1-2x_2+2 \implies -2x_1-x_2=-x_1-2x_2 \implies$. **Onto:** Let $y=\frac{x-1}{x-2} \implies y(x-2)=x-1 \implies yx-2y=x-1 \implies x(y-1)=2y-1 \implies x=\frac{2y-1}{y-1}$. Since $y\in\mathbb{R}-\{1\}$, x is real and $x\ne2$. Thus, f is bijective. **Q27.** Apply row operations $R_2\to R_2-R_1$ and $R_3\to R_3-R_1$. The determinant becomes $\begin{vmatrix}1&x&x^2\\0&y-x&y^2-x^2\\0&z-x&z^2-x^2\end{vmatrix}$. Taking $(y-x)$ from R_2 and $(z-x)$ from R_3 common, expand along C_1: $(y-x)(z-x)[(z+x)-(y+x)]=(y-x)(z-x)(z-y)=(x-y)(y-z)(z-x)$. **Q28.** Take log on both sides: $y\log x=x-y \implies y(1+\log x)=x \implies y=\frac{x}{1+\log x}$. Differentiating using quotient rule: $\frac{dy}{dx}=\frac{1(1+\log x)-x(\frac1x)}{(1+\log x)^2}=\frac{1+\log x-1}{(1+\log x)^2}=\frac{\log_e x}{(1+\log_e x)^2}$. **Q29.** Rewrite numerator: $x+3=A\frac{d}{dx}(5-4x-x^2)+B=A(-4-2x)+B$. Solving gives $A=-1/2,B=1$. The integral splits into $-\frac12\int\frac{-4-2x}{\sqrt{5-4x-x^2}}dx+\int\frac{1}{\sqrt{9-(x+2)^2}}dx$. Evaluates to $-\sqrt{5-4x-x^2}+\sin^{-1}(\frac{x+2}{3})+C$. **Q30.** Divide by $x\log x$: $\frac{dy}{dx}+\frac{1}{x\log x}y=\frac{2}{x^2}$. Integrating Factor (IF) $=e^{\int\frac{1}{x\log x}dx}=e^{\log(\log x)}=\log x$. Solution: $y\cdot\log x=\int\frac{2}{x^2}\log x\,dx$. Using integration by parts for the RHS: $2[-\frac{\log x}{x}-\frac1x]+C$. Thus, $y\log x=-\frac{2}{x}(\log x+1)+C$. **Q31.** Equation of the line through A and B is $\frac{x-3}{2}=\frac{y-4}{-3}=\frac{z-1}{5}=\lambda$. General point is $(2\lambda+3,-3\lambda+4,5\lambda+1)$. The line crosses the xy-plane where $z=0 \implies 5\lambda+1=0 \implies \lambda=-1/5$. The coordinates are $(3-2/5,4+3/5,0)=(13/5,23/5,0)$.

Q32. Let $AX=B$, where $A=\begin{bmatrix}1&-1&2\\0&2&-3\\3&-2&4\end{bmatrix}$. $|A|=1(8-6)-(-1)(0+9)+2(0-6)=2+9-12=-1\ne0$. Cofactor matrix transpose yields $adjA=\begin{bmatrix}2&0&-1\\-9&-2&3\\-6&-1&2\end{bmatrix}$. $A^{-1}=\frac{1}{-1}adjA$. $X=A^{-1}\begin{bmatrix}1\\1\\2\end{bmatrix}=-1\begin{bmatrix}2+0-2\\-9-2+6\\-6-1+4\end{bmatrix}=-1\begin{bmatrix}0\\-5\\-3\end{bmatrix}=\begin{bmatrix}0\\5\\3\end{bmatrix}$. So, $x=0,y=5,z=3$. **Q33.** Let the radius of the cylinder be r and height be h. In the sphere, $R^2=r^2+(h/2)^2 \implies r^2=R^2-h^2/4$. Volume $V=\pi r^2h=\pi(R^2-h^2/4)h$.

$V'(h) = \pi(R^2 - \frac{3h^2}{4})$. Set $V'(h) = 0 \implies h^2 = \frac{4R^2}{3} \implies h = \frac{2R}{\sqrt{3}}$. $V''(h) = \pi(-\frac{6h}{4}) < 0$, which proves maximum volume. **Q34.** Let $I = \int_0^{\pi/4} \log(1 + \tan x)\, dx$. Use property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$.
$I = \int_0^{\pi/4} \log(1 + \tan(\pi/4 - x))\, dx = \int_0^{\pi/4} \log\left(1 + \frac{1-\tan x}{1+\tan x}\right)dx = \int_0^{\pi/4} \log\left(\frac{2}{1+\tan x}\right)dx = \int_0^{\pi/4} [\log 2 - \log(1 + \tan x)]dx$. Adding both integrals: $2I = \int_0^{\pi/4} \log 2\, dx = \frac{\pi}{4} \log 2$. Hence, $I = \frac{\pi}{8} \log 2$. **Q35.** Area $A = \int_0^3 (\text{upper curve} - \text{lower curve})dx = \int_0^3 ((x^2 + 2) - x)dx$. Integrate: $[\frac{x^3}{3} + 2x - \frac{x^2}{2}]_0^3 = (9 + 6 - \frac{9}{2}) - (0) = 15 - 4.5 = 10.5$ (or $\frac{21}{2}$) sq units.

Q36. *(i)* Investment matrix $X = [x \quad y]$. Interest rate matrix $R = \begin{bmatrix} 0.05 \\ 0.07 \end{bmatrix}$. *(ii)* $XR = \text{Total Interest} \implies [x \quad y] \begin{bmatrix} 0.05 \\ 0.07 \end{bmatrix} =$. *(iii)* $0.05x + 0.07y = 1800$. Also, total investment $x + y = 30000 \implies y = 30000 - x$. Substitute: $0.05x + 0.07(30000 - x) = 1800 \implies 0.05x + 2100 - 0.07x = 1800 \implies 0.02x = 300 \implies x = 15000$. Hence, $y = 15000$. Exactly ₹ $15,000$ should be invested in each bond. **Q37.** *(i)* Square wire length $= 4x$, so square side $= x$. Circle wire length $= 28 - 4x \implies \text{radius } r = \frac{14-2x}{\pi}$. Total area $A(x) = x^2 + \pi\left(\frac{14-2x}{\pi}\right)^2 = x^2 + \frac{(14-2x)^2}{\pi}$. *(ii)* $\frac{dA}{dx} = 2x + \frac{2(14-2x)(-2)}{\pi} = 2x - \frac{56-8x}{\pi}$. *(iii)* Set $\frac{dA}{dx} = 0 \implies \pi x = 28 - 4x \implies x(\pi + 4) = 28 \implies x = \frac{28}{\pi+4}$. The exact length for the square piece is $4x = \frac{112}{\pi+4}$ $\mathbf{m}$. **Q38.** *(i)* Path 1 Cartesian: $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-3}{1}$. *(ii)* Direction vectors: $\vec{b}_1 = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b}_2 = 2\hat{i} + 3\hat{j} + \hat{k}$. Dot product $\vec{b}_1 \cdot \vec{b}_2 = (1)(2) + (-1)(3) + (1)(1) = 2 - 3 + 1 = 0$. Yes, they are orthogonal. *(iii)* Shortest Distance $= \frac{|(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$. $\vec{a}_2 - \vec{a}_1 = 3\hat{i} + 3\hat{j} + 3\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 3 & 1 \end{vmatrix} = -4\hat{i} + \hat{j} + 5\hat{k}$. Numerator dot product $= -12 + 3 + 15 = 6$. Magnitude denominator $= \sqrt{16 + 1 + 25} = \sqrt{42}$. Shortest distance $= \frac{6}{\sqrt{42}}$ (or $\frac{\sqrt{42}}{7}$) units.
